Problem Set 3

CHEM 5541 - Chemical Dynamics

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1.) Stern-Volmer Analysis

1a.)

$$\frac{d[\mathrm{Br}^*]}{dt} = -k_1[\mathrm{Br}^*][\mathrm{HCl}] - k_2[\mathrm{Br}^*][\mathrm{Br}_2]$$

$$\frac{d[\mathrm{Br}^*]}{dt} = -(k_1[\mathrm{HCl}] + k_2[\mathrm{Br}_2])[\mathrm{Br}^*]$$

Since $[\mathrm{Br}^*] << [\mathrm{Br}_2], [\mathrm{HCl}]$, the concentrations of the quenchers can be said to remain constant. Which allows us to create an effective rate constant, k_{eff}

$$\frac{d[\mathrm{Br}^*]}{dt} = -k_{eff}[\mathrm{Br}^*]$$

Which through integration:

$$[\mathrm{Br}^*(t)] = e^{-k_{eff}(t-t_0)}$$

$$[\mathrm{Br}^*(t)] = e^{(k_1[\mathrm{HCl}]+k_2[\mathrm{Br}_2])(t-t_0)}$$

1b.)

$$k_{eff} = \frac{1}{\tau} = k_1[\mathrm{HCl}] + k_2[\mathrm{Br}_2]$$

The slope of this plot is k_1 with a y-intercept of $k_2[\mathrm{Br}_2]$.

```
using Plots

plotly()
default(size=(900, 600))


R = 8.314        # Pa m^3/(mol*K)
T = 300.0        # K

P_br2_torr = [0.5, 0.5, 0.5, 0.5, 0.5]
P_br2_pas = P_br2_torr .* 133.322
conc_br2 = P_br2_pas ./ (R * T)

P_hcl_torr = [0.61, 0.97, 1.32, 2.10, 3.25]
P_hcl_pas = P_hcl_torr .* 133.322
conc_hcl = P_hcl_pas ./ (R * T)

tau_us = [1370, 1000, 750, 525, 360]
```

```
k_eff = 1 ./ (tau_us .* 1e-6)

length_conc = length(conc_hcl)
avg_conc = sum(conc_hcl) / length_conc
avg_tau = sum(k_eff) / length_conc

numerator = sum((conc_hcl .- avg_conc) .* (k_eff .- avg_tau))
denominator = sum((conc_hcl .- avg_conc) .^ 2)

k1 = numerator / denominator
intercept = avg_tau - (k1 * avg_conc)

k2 = intercept / conc_br2[1]

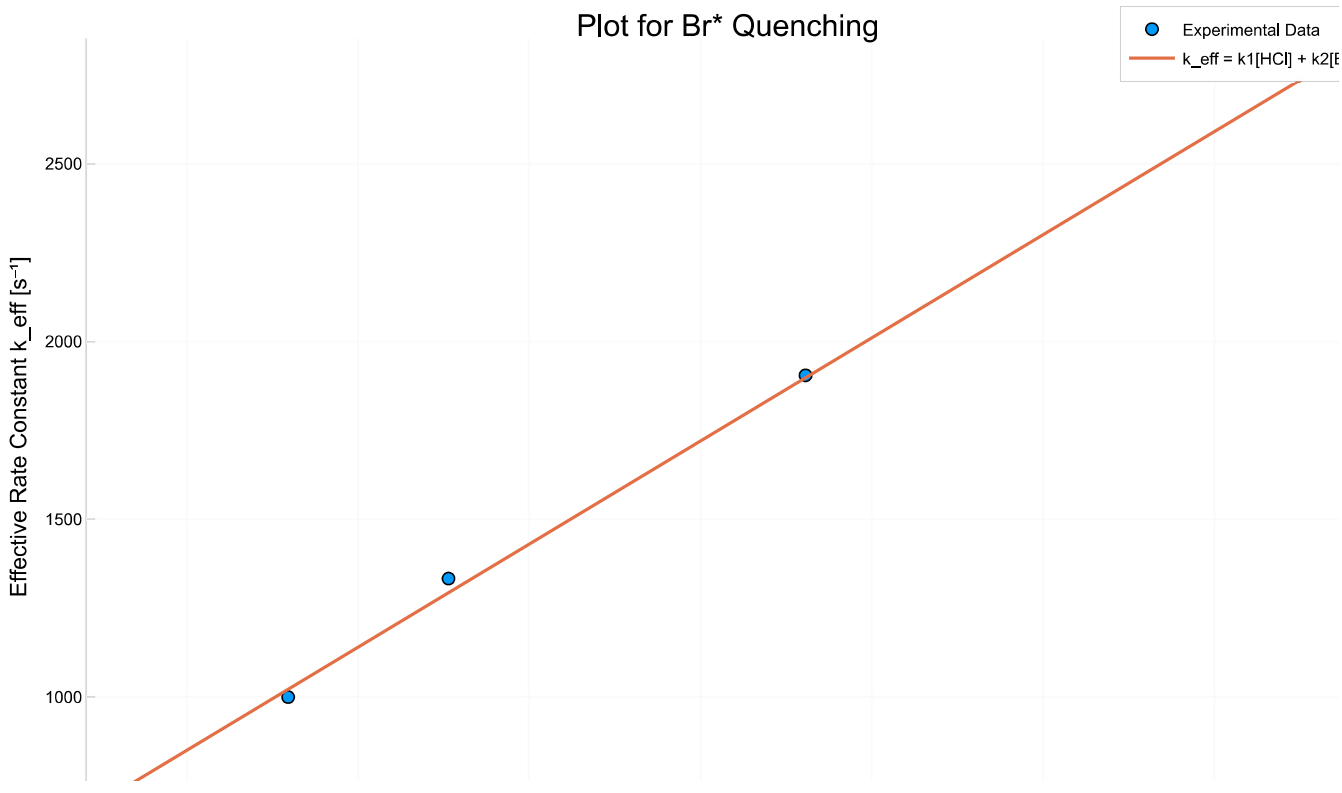
conc_range = range(minimum(conc_hcl), maximum(conc_hcl), length=100)
fit_line = k1 .* conc_range .+ intercept

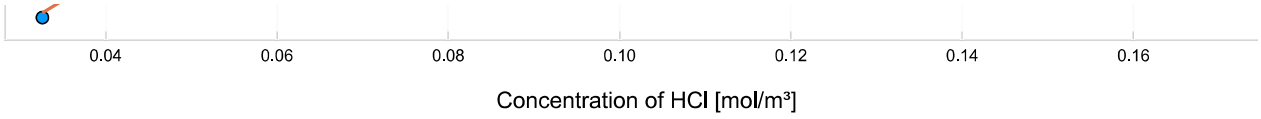
println("k1: ", round(k1, digits=2), " m³/(mol*s)")
println("k2: ", round(k2, digits=2), " m³/(mol*s)")

scatter(
    conc_hcl,
    k_eff,
    xlabel = "Concentration of HCl [mol/m³]",
    ylabel = "Effective Rate Constant k_eff [s⁻¹]",
    title = "Plot for Br* Quenching",
    label = "Experimental Data"
)

plot!(
    conc_range,
    fit_line,
    label = "k_eff = k1[HCl] + k2[Br2]",
    linewidth = 2
)
```

```
k1: 14501.01 m³/(mol*s)
k2: 10110.32 m³/(mol*s)
```





2.) Laplace Transforms

a.)

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

$$\frac{dA}{dt} = -k_1[A] \quad \longrightarrow \quad p\mathcal{L}[A] - [A]_0 = -k_1\mathcal{L}[A]$$

$$\frac{dB}{dt} = k_1[A] - k_2[B] \quad \longrightarrow \quad p\mathcal{L}[B] - [B]_0 = k_1\mathcal{L}[A] - k_2\mathcal{L}[B]$$

$$\frac{dC}{dt} = k_2[B] \quad \longrightarrow \quad p\mathcal{L}[C] - [C]_0 = k_2\mathcal{L}[B]$$

b.)

$$\mathcal{L}[A] = \frac{[A]_0}{(p + k_1)}$$

$$[A](t) = [A]_0 e^{-k_1 t}$$

c.)

$$\mathcal{L}[B] = \frac{k_1\mathcal{L}[A] + [B]_0}{p + k_2}$$

$$\mathcal{L}[B] = \frac{k_1\frac{[A]_0}{(p+k_1)} + [B]_0}{p + k_2}$$

$$\mathcal{L}[B] = \frac{k_1[A]_0 + [B]_0(p + k_1)}{(p + k_2)(p + k_1)}$$

$$\mathcal{L}[B] = \frac{k_1[A]_0}{(p + k_2)(p + k_1)} + \frac{[B]_0}{(p + k_2)}$$

$$[B](t) = \frac{k_1[A]_0}{k_2 - k_1}(e^{-k_1 t} - e^{-k_2 t}) + [B]_0 e^{-k_2 t}$$

if $[B]_0 = 0$

$$[B](t) = \frac{k_1[A]_0}{k_2 - k_1}(e^{-k_1 t} - e^{-k_2 t})$$

d.)

$$\mathcal{L}[C] = \frac{k_2\mathcal{L}[B] + [C]_0}{p}$$

$$\mathcal{L}[C] = \frac{k_1 k_2[A]_0}{p(p + k_2)(p + k_1)} + \frac{k_2[B]_0}{p(p + k_2)} + \frac{[C]_0}{p}$$

$$\mathcal{L}^{-1}\left(\frac{[C]_0}{p}\right) = [C]_0$$

$$\mathcal{L}^{-1}\left(\frac{k_2[B]_0}{p(p+k_2)}\right) = [B]_0(1-e^{-k_2 t})$$

Using partial fractions here:

$$\frac{k_1 k_2}{p(p+k_2)(p+k_1)} = \frac{X}{p} + \frac{Y}{p+k_1} + \frac{Z}{p+k_2}$$

$$X = 1 \quad Y = -\frac{k_2}{k_2-k_1} \quad Z = \frac{k_1}{k_2-k_1}$$

$$\frac{k_1 k_2}{p(p+k_2)(p+k_1)} = \frac{1}{p} + -\frac{k_2}{k_2-k_1}\frac{1}{p+k_1} + \frac{k_1}{k_2-k_1}\frac{1}{p+k_2}$$

$$\mathcal{L}^{-1} = 1 - \frac{k_2}{k_2-k_1}e^{-k_1 t} + \frac{k_1}{k_2-k_1}e^{-k_2 t}$$

$$= 1 - \frac{k_2 e^{-k_1 t} - k_2 e^{-k_2 t}}{k_2 - k_1}$$

$$= [A]_0\left[1 - \frac{k_2 e^{-k_1 t} - k_2 e^{-k_2 t}}{k_2 - k_1}\right]$$

$$[C](t) = [A]_0\left[1 - \frac{k_2 e^{-k_1 t} - k_2 e^{-k_2 t}}{k_2 - k_1}\right] + [B]_0(1-e^{-k_2 t}) + [C]_0$$

if $[B]_0 = [C]_0 = 0$

$$[C](t) = [A]_0\left[1 - \frac{k_2 e^{-k_1 t} - k_2 e^{-k_2 t}}{k_2 - k_1}\right]$$

3.) Consecutive Reactions, Pre-Equilibrium, and Rate Constants

a.)

$$A + B \rightleftharpoons AB \xrightarrow{+C} D$$

$$K = \frac{[AB]}{[A][B]}$$

$$\frac{dD}{dt} = -k[AB][C]$$

$$[AB] = K[A][B]$$

which can be subsituted into the ODE

$$\frac{dD}{dt} = kK[A][B][C]$$

with $k_r = kK$

$$R = k_r[A][B][C]$$

For the T-dependence of k_r, starting with K

$$-RT\ln K = \Delta_r H^\circ - T\Delta_r S^\circ$$

$$\ln K = -\frac{\Delta_r H^\circ}{RT} + \frac{\Delta_r S^\circ}{R}$$

$$K = e^{-\frac{\Delta_r H^\circ}{RT}} \cdot e^{\frac{\Delta_r S^\circ}{R}}$$

For k now,

$$k = Ae^{-\frac{E_a}{RT}}$$

Such that:

$$k_r = Ae^{-\frac{E_a}{RT}}\left(e^{-\frac{\Delta_r H^\circ}{RT}} \cdot e^{\frac{\Delta_r S^\circ}{R}}\right)$$

$$k_r = Ae^{\frac{\Delta_r S^\circ}{R}}\left(e^{-\frac{E_a+\Delta_r H^\circ}{RT}}\right)$$

Resulting in:

$$a = Ae^{\frac{\Delta_r S^\circ}{R}}$$

and:

$$k_r = a \cdot e^{-\frac{E_a+\Delta_r H^\circ}{RT}}$$

b.)

$$A + B \xrightleftharpoons[k_2]{k_1} C \xrightarrow{k_3} D$$

$$\frac{dD}{dt} = k_3[C]$$

Using SSA for $\frac{dC}{dt} = 0$

$$\frac{dC}{dt} = k_1[A][B] - k_2[C] - k_3[C] = 0$$

$$k_1[A][B] = [C](k_2 + k_3)$$

$$[C] = \frac{k_1[A][B]}{(k_2 + k_3)}$$

Substituting back into the expression of $\frac{dD}{dt}$:

$$\frac{dD}{dt} = \frac{k_3 k_1[A][B]}{(k_2 + k_3)}$$

Defining $k_r \equiv \frac{k_3 k_1}{(k_2+k_3)}$ to get to the psuedo-second-order rate law:

$$\frac{dD}{dt} = k_r[A][B]$$

If $k_3 \ll k_2$

$$k_r \approx \frac{k_3 k_1}{k_2}$$

$$k_r \approx \frac{A_3 e^{-\frac{E_{a,3}}{RT}} A_1 e^{-\frac{E_{a,1}}{RT}}}{A_2 e^{-\frac{E_{a,2}}{RT}}}$$

$$k_r \approx \frac{A_3 A_1}{A_2}\left(e^{-\frac{E_{a,3}+E_{a,1}-E_{a,2}}{RT}}\right)$$

$$A_r = \frac{A_3 A_1}{A_2}$$

$$E_{a,r} = E_{a,3} + E_{a,1} - E_{a,2}$$

4.) Unimolecular Reaction with Autocatalysis

a.)

$$\frac{dA}{dt} = -k_1[A] - k_2[A][B]$$

$$\frac{dB}{dt} = k_1[A] + k_2[A][B]$$

b.)

$$N = A_0 + B_0$$

The total moles must also be equal when changing, so:

$$\frac{dN}{dt} = \frac{dA}{dt} + \frac{dB}{dt}$$

$$\frac{dN}{dt} = (-k_1[A] - k_2[A][B]) + (k_1[A] + k_2[A][B]) = 0$$

Which further shows that the total moles throughout the reaction does not change.

c.)

$$\frac{dA}{dt} = rA\left(1 - \frac{A}{K}\right)$$

Need the above equation in terms of k_1, k_2, A_0, B_0. A is in the equation, but not B, so subsitute that.

$$\frac{dA}{dt} = -k_1[A] - k_2[A]\left[([A]_0 + [B]_0) - [A]\right]$$

$$\frac{dA}{dt} = -k_1[A] - k_2[A]([A]_0 + [B]_0) + k_2[A]^2$$

$$\frac{dA}{dt} = -(k_1 + k_2([A]_0 + [B]_0))[A] + k_2[A]^2$$

$$\frac{dA}{dt} = -[k_1 + k_2(A_0 + B_0)]A\left(1 - \frac{k_2 A}{k_1 + k_2(A_0 + B_0)}\right)$$

Which is of the same form as the original equation such that:

$$r \equiv -[k_1 + k_2(A_0 + B_0)]$$

$$K \equiv \frac{k_1 + k_2(A_0 + B_0)}{k_2}$$

d.)

$$A(t) = \frac{KA_0e^{rt}}{K + A_0(e^{rt} - 1)}$$

$$A(t) = \frac{\frac{k_1+k_2(A_0+B_0)}{k_2} A_0 e^{-[k_1+k_2(A_0+B_0)]t}}{\frac{k_1+k_2(A_0+B_0)}{k_2} + A_0(e^{-[k_1+k_2(A_0+B_0)]t} - 1)}$$

```
using Plots

plotly()

t_range = range(0, 10, length=100)

k_1 = 0.01      # s^-1
k_2 = 1         # L mol^-1 s^-1
A_0 = 1         # mol/L
B_0 = 0
k_1_comp = 1    # s^-1

K = (k_1 + k_2 * (A_0 + B_0)) / k_2

r = -(k_1 + k_2 * (A_0 + B_0))

A_t = (K * A_0 * exp.(r * t_range)) ./ (K .+ A_0 * (exp.(r * t_range) .- 1))

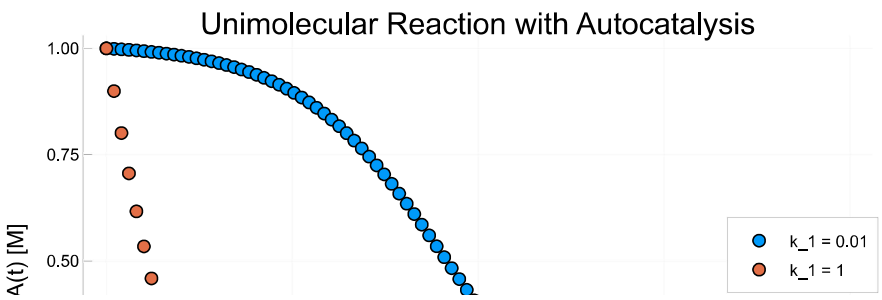
K_comp = (k_1_comp + k_2 * (A_0 + B_0)) / k_2

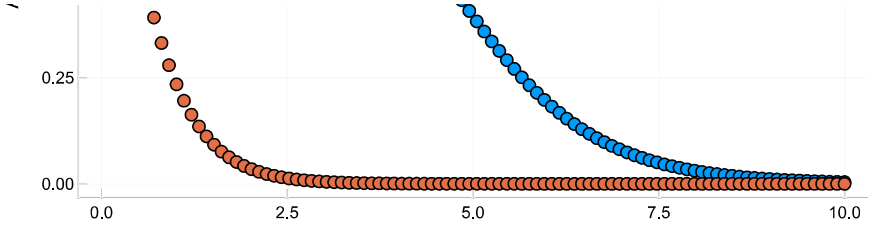
r_comp = -(k_1_comp + k_2 * (A_0 + B_0))

A_t_comp = (K_comp * A_0 * exp.(r_comp * t_range)) ./ (K_comp .+ A_0 * (exp.(r_comp * t_range) .- 1))

scatter(
    t_range,
    A_t,
    label="k_1 = 0.01",
    xlabel="Time [s]",
    ylabel="A(t) [M]",
    title="Unimolecular Reaction with Autocatalysis",
    legend=:right
)

scatter!(
    t_range,
    A_t_comp,
    label="k_1 = 1",
    xlabel="Time [s]",
    ylabel="A(t) [M]"
)
```





When $k_1 \ll k_2$, the autocatalytic pathway dominates and causes a slow decline in $[A]$ as $[B]$ builds up, falling off once $[B]$ is sufficiently high to catalyze enough reactions. When $k_1 = k_2$, the unimolecular route dominates and causes a rapid decline in $[A]$ as the slower B catalyzed pathway does not gain sufficient momentum.

5.) Simulating Consecutive Reactions

a.)

$$A \xrightleftharpoons[k_{A1}]{k_{A2}} B \xrightleftharpoons[k_{B1}]{k_{B2}} C$$

```
using CSV
using DataFrames
using Plots

function kinetics_simulation()
    A_0 = 1.0         # mol/L
    B_0 = 0.0         # mol/L
    C_0 = 0.0         # mol/L
    k_a_1 = 0.2       # s^-1
    k_a_2 = 0.2       # s^-1
    k_b_1 = 0.2       # s^-1
    k_b_2 = 0.2       # s^-1
    delta_t = 1.0     # s
    T = 50            # s


    t_array = []
    A_array = []
    B_array = []
    C_array = []

    A = A_0
    B = B_0
    C = C_0
    t = 0.0

    open("S.Arnold_PS3_Q5ai_output.txt", "w") do io
        write(io, "# Parameters: A_0=$A_0 mol/L, B_0=$B_0 mol/L, C_0=$C_0 mol/L, k_a_1=$k_a_1 s^-1, k_a_
        write(io, "Time(s)\t[A](mol/L)\t[B](mol/L)\t[C](mol/L)\n")

        while t <= T
            push!(t_array, t)
            push!(A_array, A)
            push!(B_array, B)
            push!(C_array, C)

            write(io, "$t\t$A\t$B\t$C\n")

            # Rate laws
            dA = (-k_a_1 * A + k_a_2 * B) * delta_t
            dB = (k_a_1 * A + k_b_2 * C - k_b_1 * B - k_a_2 * B) * delta_t
```

```
            dC = (-k_b_2 * C + k_b_1 * B) * delta_t

            A += dA
            B += dB
            C += dC
            t += delta_t
        end
    end

    df = CSV.read("S.Arnold_PS3_Q5ai_output.txt", DataFrame, delim='\t', comment="#")
    print(df)

    Plots.plot(
        t_array,
        A_array,
        title="Consecutive Reactions Kinetics Simulation (All k's Equal)",
        label="[A]",
        xlabel="Time [s]",
        ylabel="Concentration [mol/L]",
        color=:blue,
        linewidth=2,
        legend=:outertopright
    )

    plot!(
        t_array,
        B_array,
        label="[B]",
        color=:red,
        linewidth=2
    )

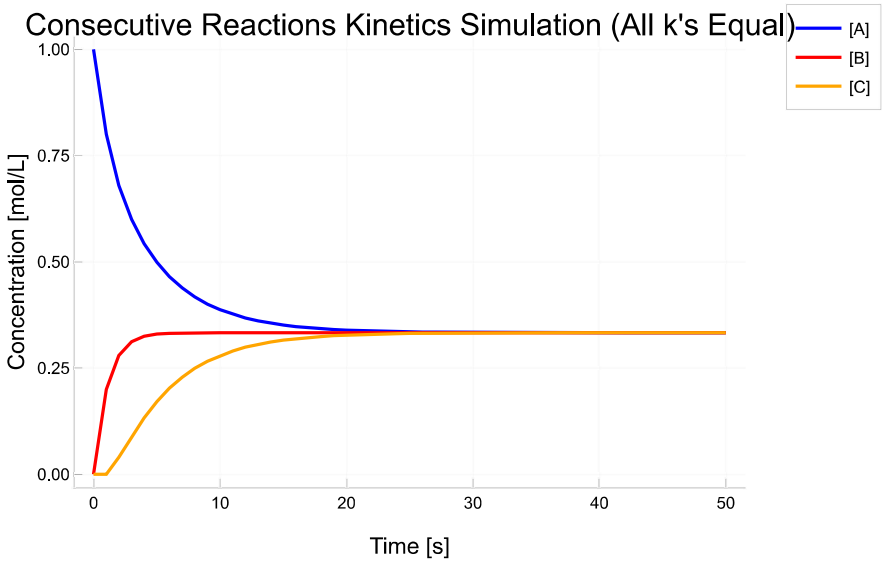
    plot!(
        t_array,
        C_array,
        label="[C]",
        color=:orange,
        linewidth=2
    )

end

kinetics_simulation()
```

```
51×4 DataFrame
 Row │ Time(s)  [A](mol/L)  [B](mol/L)  [C](mol/L)
     │ Float64  Float64     Float64     Float64
─────┼──────────────────────────────────────────────
   1 │     0.0    1.0         0.0         0.0
   2 │     1.0    0.8         0.2         0.0
   3 │     2.0    0.68        0.28        0.04
   4 │     3.0    0.6         0.312       0.088
   5 │     4.0    0.5424      0.3248      0.1328
   6 │     5.0    0.49888     0.32992     0.1712
   7 │     6.0    0.465088    0.331968    0.202944
   8 │     7.0    0.438464    0.332787    0.228749
   9 │     8.0    0.417329    0.333115    0.249556
  10 │     9.0    0.400486    0.333246    0.266268
  11 │    10.0    0.387038    0.333298    0.279664
  12 │    11.0    0.37629     0.333319    0.290391
  13 │    12.0    0.367696    0.333328    0.298976
  14 │    13.0    0.360822    0.333331    0.305847
  15 │    14.0    0.355324    0.333332    0.311344
  16 │    15.0    0.350926    0.333333    0.315741
  17 │    16.0    0.347407    0.333333    0.31926
```

18	17.0	0.344592	0.333333	0.322074
19	18.0	0.342341	0.333333	0.324326
20	19.0	0.340539	0.333333	0.326128
21	20.0	0.339098	0.333333	0.327569
22	21.0	0.337945	0.333333	0.328722
23	22.0	0.337023	0.333333	0.329644
24	23.0	0.336285	0.333333	0.330382
25	24.0	0.335695	0.333333	0.330972
26	25.0	0.335222	0.333333	0.331444
27	26.0	0.334844	0.333333	0.331822
28	27.0	0.334542	0.333333	0.332124
29	28.0	0.3343	0.333333	0.332366
30	29.0	0.334107	0.333333	0.33256
31	30.0	0.333952	0.333333	0.332714
32	31.0	0.333829	0.333333	0.332838
33	32.0	0.333729	0.333333	0.332937
34	33.0	0.33365	0.333333	0.333016
35	34.0	0.333587	0.333333	0.33308
36	35.0	0.333536	0.333333	0.333131
37	36.0	0.333496	0.333333	0.333171
38	37.0	0.333463	0.333333	0.333204
39	38.0	0.333437	0.333333	0.333229
40	39.0	0.333416	0.333333	0.33325
41	40.0	0.3334	0.333333	0.333267
42	41.0	0.333387	0.333333	0.33328
43	42.0	0.333376	0.333333	0.333291
44	43.0	0.333367	0.333333	0.333299
45	44.0	0.333361	0.333333	0.333306
46	45.0	0.333355	0.333333	0.333312
47	46.0	0.333351	0.333333	0.333316
48	47.0	0.333347	0.333333	0.333319
49	48.0	0.333344	0.333333	0.333322
50	49.0	0.333342	0.333333	0.333324
51	50.0	0.33334	0.333333	0.333326



b.)

```
using CSV
using DataFrames
using Plots

function kinetics_simulation()
    A_0 = 1.0          # mol/L
    B_0 = 0.0          # mol/L
    C_0 = 0.0          # mol/L
```

```
k_a_1 = 0.1       # s^-1
k_a_2 = 0.0       # s^-1
k_b_1 = 0.1       # s^-1
k_b_2 = 0.0       # s^-1
delta_t = 0.1     # s
T = 50            # s


t_array = []
A_array = []
B_array = []
C_array = []

A = A_0
B = B_0
C = C_0
t = 0.0

open("S.Arnold_PS3_Q5aii_output.txt", "w") do io
    write(io, "# Parameters: A_0=$A_0 mol/L, B_0=$B_0 mol/L, C_0=$C_0 mol/L, k_a_1=$k_a_1 s^-1, k_a_
    write(io, "Time(s)\t[A](mol/L)\t[B](mol/L)\t[C](mol/L)\n")

    while t <= T
        push!(t_array, t)
        push!(A_array, A)
        push!(B_array, B)
        push!(C_array, C)

        write(io, "$t\t$A\t$B\t$C\n")

        # Rate laws
        dA = (-k_a_1 * A + k_a_2 * B) * delta_t
        dB = (k_a_1 * A + k_b_2 * C - k_b_1 * B - k_a_2 * B) * delta_t
        dC = (-k_b_2 * C + k_b_1 * B) * delta_t

        A += dA
        B += dB
        C += dC
        t += delta_t
    end
end

df = CSV.read("S.Arnold_PS3_Q5aii_output.txt", DataFrame, delim='\t', comment="#")
print(df, "\n")

B_max_row = argmax(df[!, :"[B](mol/L)"])

max_time = df[B_max_row, :"Time(s)"]
max_val_B = df[B_max_row, :"[B](mol/L)"]

println("-"^50)
println("\nMaximum of B is ", round(max_val_B, digits=4), " mol/L at Time: ", round(max_time, digits
println("-"^50)

Plots.plot(
    t_array,
    A_array,
    title="Consecutive Reactions Kinetics Simulation (Only Forward)",
    label="[A]",
    xlabel="Time [s]",
    ylabel="Concentration [mol/L]",
    color=:blue,
    linewidth=2,
    legend=:outertopright
)
```

```
    plot!(
        t_array,
        B_array,
        label="[B]",
        color=:red,
        linewidth=2
    )

    plot!(
        t_array,
        C_array,
        label="[C]",
        color=:orange,
        linewidth=2
    )

end

kinetics_simulation()
```

500×4 DataFrame

Row	Time(s)	[A](mol/L)	[B](mol/L)	[C](mol/L)
	Float64	Float64	Float64	Float64
1	0.0	1.0	0.0	0.0
2	0.1	0.99	0.01	0.0
3	0.2	0.9801	0.0198	0.0001
4	0.3	0.970299	0.029403	0.000298
5	0.4	0.960596	0.038812	0.00059203
6	0.5	0.95099	0.0480298	0.00098015
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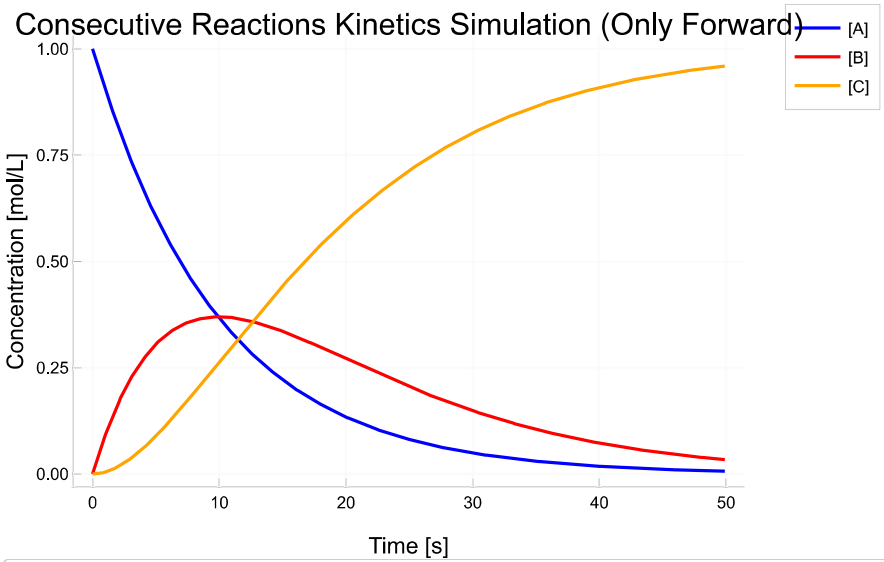
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339 |    33.8  0.0334731    0.114282   0.852245
340 |    33.9  0.0331384    0.113474   0.853388
341 |    34.0  0.032807     0.11267    0.854523
342 |    34.1  0.0324789    0.111872   0.855649
343 |    34.2  0.0321541    0.111078   0.856768
344 |    34.3  0.0318326    0.110289   0.857879
345 |    34.4  0.0315142    0.109504   0.858982
346 |    34.5  0.0311991    0.108724   0.860077
347 |    34.6  0.0308871    0.107949   0.861164
348 |    34.7  0.0305782    0.107178   0.862243
349 |    34.8  0.0302725    0.106412   0.863315
350 |    34.9  0.0299697    0.105651   0.864379
351 |    35.0  0.02967      0.104894   0.865436
352 |    35.1  0.0293733    0.104142   0.866485
353 |    35.2  0.0290796    0.103394   0.867526
354 |    35.3  0.0287888    0.102651   0.86856
355 |    35.4  0.0285009    0.101912   0.869587
356 |    35.5  0.0282159    0.101178   0.870606
357 |    35.6  0.0279338    0.100449   0.871618
358 |    35.7  0.0276544    0.0997235  0.872622
359 |    35.8  0.0273779    0.0990028  0.873619
360 |    35.9  0.0271041    0.0982866  0.874609
361 |    36.0  0.0268331    0.0975747  0.875592
362 |    36.1  0.0265647    0.0968673  0.876568
363 |    36.2  0.0262991    0.0961643  0.877537
364 |    36.3  0.0260361    0.0954656  0.878498
365 |    36.4  0.0257757    0.0947713  0.879453
366 |    36.5  0.025518     0.0940814  0.880401
367 |    36.6  0.0252628    0.0933957  0.881341
368 |    36.7  0.0250102    0.0927144  0.882275
369 |    36.8  0.0247601    0.0920374  0.883203
370 |    36.9  0.0245125    0.0913646  0.884123
371 |    37.0  0.0242673    0.0906961  0.885037
```

```
372 |    37.1  0.0240247    0.0900318  0.885944
373 |    37.2  0.0237844    0.0893717  0.886844
374 |    37.3  0.0235466    0.0887159  0.887738
375 |    37.4  0.0233111    0.0880642  0.888625
376 |    37.5  0.023078     0.0874166  0.889505
377 |    37.6  0.0228472    0.0867732  0.89038
378 |    37.7  0.0226187    0.086134   0.891247
379 |    37.8  0.0223926    0.0854988  0.892109
380 |    37.9  0.0221686    0.0848678  0.892964
381 |    38.0  0.0219469    0.0842408  0.893812
382 |    38.1  0.0217275    0.0836178  0.894655
383 |    38.2  0.0215102    0.0829989  0.895491
384 |    38.3  0.0212951    0.082384   0.896321
385 |    38.4  0.0210821    0.0817732  0.897145
386 |    38.5  0.0208713    0.0811662  0.897962
387 |    38.6  0.0206626    0.0805633  0.898774
388 |    38.7  0.020456     0.0799643  0.89958
389 |    38.8  0.0202514    0.0793692  0.900379
390 |    38.9  0.0200489    0.078778   0.901173
391 |    39.0  0.0198484    0.0781907  0.901961
392 |    39.1  0.0196499    0.0776073  0.902743
393 |    39.2  0.0194534    0.0770277  0.903519
394 |    39.3  0.0192589    0.076452   0.904289
395 |    39.4  0.0190663    0.0758801  0.905054
396 |    39.5  0.0188756    0.0753119  0.905812
397 |    39.6  0.0186869    0.0747476  0.906566
398 |    39.7  0.0185       0.074187   0.907313
399 |    39.8  0.018315     0.0736301  0.908055
400 |    39.9  0.0181319    0.0730769  0.908791
401 |    40.0  0.0179506    0.0725275  0.909522
402 |    40.1  0.017771     0.0719817  0.910247
403 |    40.2  0.0175933    0.0714396  0.910967
404 |    40.3  0.0174174    0.0709011  0.911681
405 |    40.4  0.0172432    0.0703663  0.91239
406 |    40.5  0.0170708    0.0698351  0.913094
407 |    40.6  0.0169001    0.0693074  0.913792
408 |    40.7  0.0167311    0.0687834  0.914486
409 |    40.8  0.0165638    0.0682628  0.915173
410 |    40.9  0.0163981    0.0677459  0.915856
411 |    41.0  0.0162342    0.0672324  0.916533
412 |    41.1  0.0160718    0.0667224  0.917206
413 |    41.2  0.0159111    0.0662159  0.917873
414 |    41.3  0.015752     0.0657128  0.918535
415 |    41.4  0.0155945    0.0652132  0.919192
416 |    41.5  0.0154385    0.064717   0.919844
417 |    41.6  0.0152841    0.0642243  0.920492
418 |    41.7  0.0151313    0.0637349  0.921134
419 |    41.8  0.01498      0.0632488  0.921771
420 |    41.9  0.0148302    0.0627661  0.922404
421 |    42.0  0.0146819    0.0622868  0.923031
422 |    42.1  0.0145351    0.0618107  0.923654
423 |    42.2  0.0143897    0.061338   0.924272
424 |    42.3  0.0142458    0.0608685  0.924886
425 |    42.4  0.0141034    0.0604023  0.925494
426 |    42.5  0.0139623    0.0599393  0.926098
427 |    42.6  0.0138227    0.0594795  0.926698
428 |    42.7  0.0136845    0.0590229  0.927293
429 |    42.8  0.0135476    0.0585695  0.927883
430 |    42.9  0.0134122    0.0581193  0.928469
431 |    43.0  0.013278     0.0576723  0.92905
432 |    43.1  0.0131453    0.0572283  0.929626
433 |    43.2  0.0130138    0.0567875  0.930199
434 |    43.3  0.0128837    0.0563497  0.930767
435 |    43.4  0.0127548    0.0559151  0.93133
436 |    43.5  0.0126273    0.0554835  0.931889
437 |    43.6  0.012501     0.0550549  0.932444
```

```
438 |    43.7  0.012376     0.0546294  0.932995
439 |    43.8  0.0122522    0.0542068  0.933541
440 |    43.9  0.0121297    0.0537873  0.934083
441 |    44.0  0.0120084    0.0533707  0.934621
442 |    44.1  0.0118883    0.0529571  0.935155
443 |    44.2  0.0117694    0.0525464  0.935684
444 |    44.3  0.0116518    0.0521386  0.93621
445 |    44.4  0.0115352    0.0517338  0.936731
446 |    44.5  0.0114199    0.0513318  0.937248
447 |    44.6  0.0113057    0.0509327  0.937762
448 |    44.7  0.0111926    0.0505364  0.938271
449 |    44.8  0.0110807    0.050143   0.938776
450 |    44.9  0.0109699    0.0497523  0.939278
451 |    45.0  0.0108602    0.0493645  0.939775
452 |    45.1  0.0107516    0.0489795  0.940269
453 |    45.2  0.0106441    0.0485972  0.940759
454 |    45.3  0.0105376    0.0482177  0.941245
455 |    45.4  0.0104323    0.0478409  0.941727
456 |    45.5  0.0103279    0.0474668  0.942205
457 |    45.6  0.0102247    0.0470954  0.94268
458 |    45.7  0.0101224    0.0467267  0.943151
459 |    45.8  0.0100212    0.0463606  0.943618
460 |    45.9  0.00992097   0.0459972  0.944082
461 |    46.0  0.00982176   0.0456365  0.944542
462 |    46.1  0.00972355   0.0452783  0.944998
463 |    46.2  0.00962631   0.0449228  0.945451
464 |    46.3  0.00953005   0.0445698  0.9459
465 |    46.4  0.00943475   0.0442194  0.946346
466 |    46.5  0.0093404    0.0438716  0.946788
467 |    46.6  0.009247     0.0435263  0.947227
468 |    46.7  0.00915453   0.0431835  0.947662
469 |    46.8  0.00906298   0.0428432  0.948094
470 |    46.9  0.00897235   0.0425054  0.948522
471 |    47.0  0.00888263   0.0421701  0.948947
472 |    47.1  0.0087938    0.0418372  0.949369
473 |    47.2  0.00870586   0.0415067  0.949787
474 |    47.3  0.0086188    0.0411787  0.950202
475 |    47.4  0.00853262   0.0408531  0.950614
476 |    47.5  0.00844729   0.0405299  0.951023
477 |    47.6  0.00836282   0.0402091  0.951428
478 |    47.7  0.00827919   0.0398906  0.95183
479 |    47.8  0.0081964    0.0395745  0.952229
480 |    47.9  0.00811443   0.0392607  0.952625
481 |    48.0  0.00803329   0.0389493  0.953017
482 |    48.1  0.00795296   0.0386401  0.953407
483 |    48.2  0.00787343   0.0383332  0.953793
484 |    48.3  0.00779469   0.0380287  0.954177
485 |    48.4  0.00771675   0.0377263  0.954557
486 |    48.5  0.00763958   0.0374262  0.954934
487 |    48.6  0.00756318   0.0371283  0.955308
488 |    48.7  0.00748755   0.0368327  0.95568
489 |    48.8  0.00741268   0.0365392  0.956048
490 |    48.9  0.00733855   0.036248   0.956413
491 |    49.0  0.00726516   0.0359589  0.956776
492 |    49.1  0.00719251   0.0356719  0.957136
493 |    49.2  0.00712059   0.0353872  0.957492
494 |    49.3  0.00704938   0.0351045  0.957846
495 |    49.4  0.00697889   0.0348239  0.958197
496 |    49.5  0.0069091    0.0345455  0.958545
497 |    49.6  0.00684001   0.0342691  0.958891
498 |    49.7  0.00677161   0.0339948  0.959234
499 |    49.8  0.00670389   0.0337226  0.959574
500 |    49.9  0.00663685   0.0334524  0.959911
-------------------------------------------------

Maximum of B is 0.3697 mol/L at Time: 9.9 s
```

```
------------------------------------------------
```




```
using CSV
using DataFrames
using Plots

function kinetics_simulation()
    A_0 = 1.0          # mol/L
    B_0 = 0.0          # mol/L
    C_0 = 0.0          # mol/L
    k_a_1 = 0.1        # s^-1
    k_a_2 = 0.0        # s^-1
    k_b_1 = 0.1        # s^-1
    k_b_2 = 0.0        # s^-1
    delta_t = 0.1      # s
    T = 50             # s
    delta_k = 0.1    # s^-1
    num_steps = 5

    A_colors = palette(:blues, rev = true, num_steps)
    B_colors = palette(:reds, rev = true, num_steps)
    C_colors = palette(:autumn1, rev = true, num_steps)

    initial_plot = Plots.plot(
            title="Consecutive Reactions Kinetics Simulation (Increasing k)",
            xlabel="Time [s]",
            ylabel="Concentration [mol/L]",
            legend=:outerright
    )

    results_str = "\n" * "="^45 * "\n"
    results_str *= rpad("k (s^-1)", 10) * " | " * rpad("Max [B]", 12) * " | " * rpad("t_B,max (s)", 15)
    results_str *= "-"^45 * "\n"

    open("S.Arnold_PS3_Q5aiii_output.txt", "w") do io
        write(io, "# Parameters: A_0=$A_0 mol/L, B_0=$B_0 mol/L, C_0=$C_0 mol/L, k_a_1=$k_a_1 s^-1, k_a_
        write(io, "Time(s)\t[A](mol/L)\t[B](mol/L)\t[C](mol/L)\n")


        for step in 0:(num_steps-1)
            t_array = []
            A_array = []
            B_array = []
            C_array = []
```

```
A = A_0
B = B_0
C = C_0
t = 0.0
k_a_1_it = k_a_1 + step * delta_k
# k_b_1_it = k_b_1 + step * delta_k

while t <= T
    push!(t_array, t)
    push!(A_array, A)
    push!(B_array, B)
    push!(C_array, C)

    write(io, "$step\t$t\t$A\t$B\t$C\n")

    # Rate laws
    dA = (-k_a_1_it * A + k_a_2 * B) * delta_t
    dB = (k_a_1_it * A + k_b_2 * C - k_b_1 * B - k_a_2 * B) * delta_t
    dC = (-k_b_2 * C + k_b_1 * B) * delta_t

    A += dA
    B += dB
    C += dC
    t += delta_t
end

max_idx = argmax(B_array)
max_val_B = B_array[max_idx]
max_time = t_array[max_idx]

lbl_k = round(k_a_1_it, digits=2)

k_str = rpad(string(round(lbl_k, digits=2)), 10)
b_str = rpad(string(round(max_val_B, digits=4)), 12)
t_str = rpad(string(round(max_time, digits=2)), 15)
results_str *= k_str * " | " * b_str * " | " * t_str * "\n"

current_A_color = A_colors[step + 1]
current_B_color = B_colors[step + 1]
current_C_color = C_colors[step + 1]

plot!(
    initial_plot,
    t_array,
    A_array,
    label="",
    color=current_A_color,
    linewidth=2,
)

plot!(
    initial_plot,
    t_array,
    B_array,
    label="[B] (k=$lbl_k)",
    color=current_B_color,
    linewidth=2,
)

plot!(
    initial_plot,
    t_array,
    C_array,
    label="",
    color=current_C_color,
    linewidth=2,
```

```
            )

            scatter!(
                initial_plot,
                [max_time],
                [max_val_B],
                color=current_B_color,
                label="",
                markersize=4
            )
        end
    end

    results_str *= "="^45 * "\n"

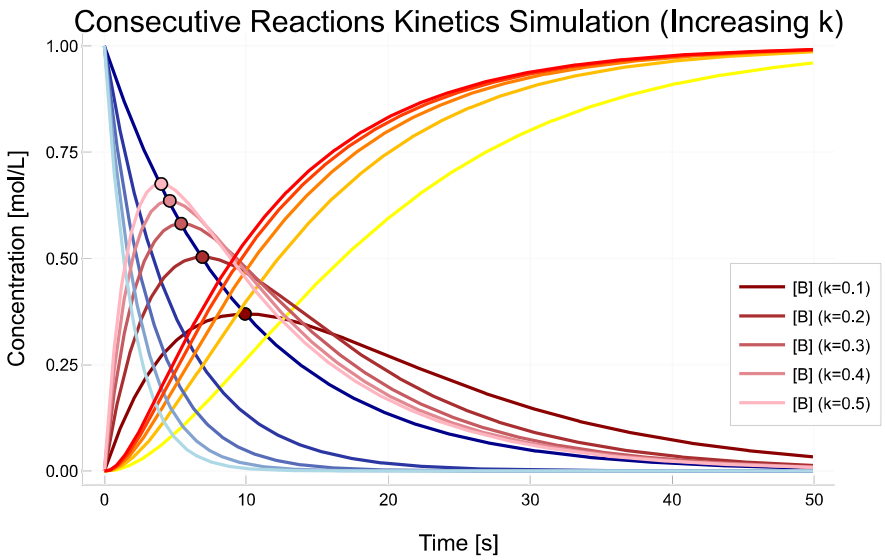
    println(results_str)
    flush(stdout)

    return initial_plot

end

kinetics_simulation()
```

```
=============================================
k (s^-1)    | Max [B]       | t_B,max (s)
---------------------------------------------
0.1         | 0.3697        | 9.9
0.2         | 0.5035        | 6.9
0.3         | 0.5822        | 5.4
0.4         | 0.6359        | 4.6
0.5         | 0.6756        | 4.0
=============================================
```



C.)

$$B = \frac{k_a}{k_b - k_a}(e^{-k_a t} - e^{-k_b t})[A]_0$$

$$\frac{dB}{dt} = \frac{k_a[A]_0}{k_b - k_a}(-k_a e^{-k_a t} + k_b e^{-k_b t}) = 0$$

$$\left(\frac{k_b}{k_a}\right) = e^{(k_b - k_a)t}$$

$$t_{max} = \frac{\ln\left(\frac{k_b}{k_a}\right)}{(k_b - k_a)}$$

$$B_{max} = \frac{k_a[A]_0}{k_b - k_a}(e^{-k_a t_{max}} - e^{-k_b t_{max}})$$

```
using CSV
using DataFrames
using Plots

function kinetics_simulation()
    A_0 = 1.0         # mol/L
    B_0 = 0.0         # mol/L
    C_0 = 0.0         # mol/L
    k_a_1 = 0.1       # s^-1
    k_a_2 = 0.0       # s^-1
    k_b_1 = 0.1       # s^-1
    k_b_2 = 0.0       # s^-1
    delta_t = 0.1    # s
    T = 50            # s
    delta_k = 0.1     # s^-1
    num_steps = 5


    k_a_1_points = []
    t_max_sim = []
    t_max_calc = []

    results_str = "\n" * "="^70 * "\n"
    results_str *= rpad("k_A1", 8) * " | " * rpad("t_max(Sim)", 12) * " | " * rpad("t_max(Calc)", 12) *
    results_str *= "-"^70 * "\n"

    for step in 0:(num_steps-1)
        t_array = []
        B_array = []

        A = A_0
        B = B_0;
        C = C_0
        t = 0.0
        k_a_1_it = k_a_1 + step * delta_k

        while t <= T
            push!(t_array, t)
            push!(B_array, B)

            dA = (-k_a_1_it * A + k_a_2 * B) * delta_t
            dB = (k_a_1_it * A + k_b_2 * C - k_b_1 * B - k_a_2 * B) * delta_t
            dC = (-k_b_2 * C + k_b_1 * B) * delta_t

            A += dA;
            B += dB;
            C += dC;
            t += delta_t
        end

        max_idx = argmax(B_array)
        max_val_B = B_array[max_idx]
        max_time = t_array[max_idx]

        calc_t = 0.0
        calc_B = 0.0
```

```
        if step == 0
            calc_t = 1.0 / k_b_1
            calc_B = A_0 * exp(-1.0)
        else
            calc_t = log(k_b_1 / k_a_1_it) / (k_b_1 - k_a_1_it)
            calc_B = A_0 * (k_a_1_it / k_b_1)^(k_b_1 / (k_b_1 - k_a_1_it))
        end

        push!(k_a_1_points, k_a_1_it)
        push!(t_max_sim, max_time)
        push!(t_max_calc, calc_t)

        r_ka   = rpad(string(round(k_a_1_it, digits=1)), 8)
        r_tsim = rpad(string(round(max_time, digits=2)), 12)
        r_tcal = rpad(string(round(calc_t, digits=2)), 12)
        r_bsim = rpad(string(round(max_val_B, digits=4)), 12)
        r_bcal = rpad(string(round(calc_B, digits=4)), 12)

        results_str *= r_ka * " | " * r_tsim * " | " * r_tcal * " | " * r_bsim * " | " * r_bcal * "\n"
    end
    results_str *= "="^70 * "\n"

    println(results_str)
    flush(stdout)

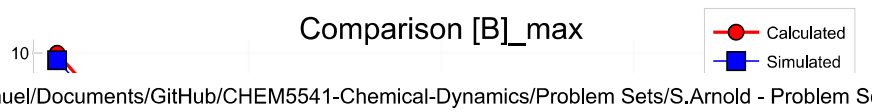
    plot(
        k_a_1_points,
        t_max_calc,
        label="Calculated",
        linewidth=2,
        color=:red,
        xlabel="k_A_1 [s⁻¹]",
        ylabel=" t_B,max [s]",
        title="Comparison [B]_max",
        marker=:circle,
        markersize=5
    )

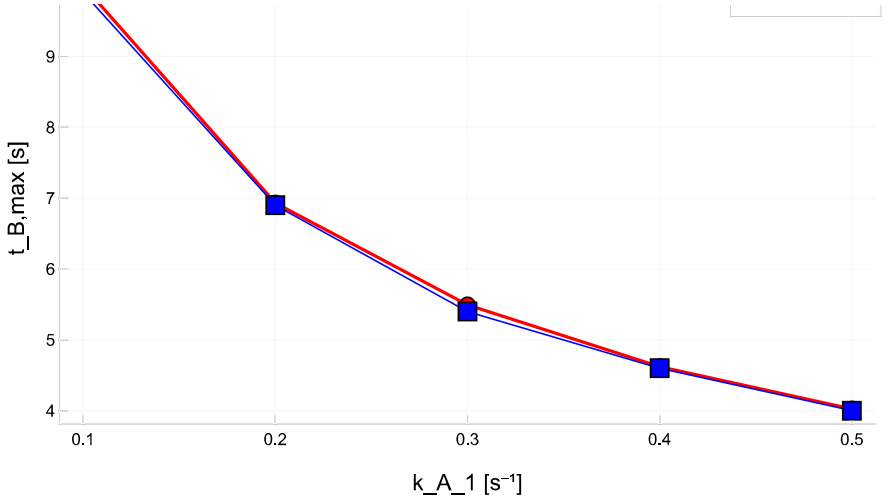
    plot!(
        k_a_1_points,
        t_max_sim,
        label="Simulated",
        color=:blue,
        markersize=6,
        marker=:square
    )

end

kinetics_simulation()
```

```
======================================================================
k_A1     | t_max(Sim)   | t_max(Calc)  | B_max(Sim)   | B_max(Calc)
----------------------------------------------------------------------
0.1      | 9.9          | 10.0         | 0.3697       | 0.3679
0.2      | 6.9          | 6.93         | 0.5035       | 0.5
0.3      | 5.4          | 5.49         | 0.5822       | 0.5774
0.4      | 4.6          | 4.62         | 0.6359       | 0.63
0.5      | 4.0          | 4.02         | 0.6756       | 0.6687
======================================================================
```





If $k_{a_1} \rightarrow k_{b_1}$

$$B = \frac{k_a[A]_0}{k_b - k_a}(e^{-k_a t} - e^{-k_b t}) = \frac{0}{0}$$

So we define a tiny difference that shrinks to zero as $k_{a_1} \rightarrow k_{b_1}$ $k_{a_1} = k_{b_1} + \Delta k$

$$e^{-k_{a_1} t} = e^{-(k_{b_1} + \Delta k)t} = e^{-k_{b_1}} e^{-\Delta k t}$$

Using a Taylor series expansion on $e^{-\Delta k t}$, $(e^x = 1 + x + \frac{x^2}{2!} + \dots)$

$$e^{-\Delta k t} \approx 1 - (\Delta k t) + \frac{(\Delta k t)^2}{2} + \dots$$

First two terms are enough given that Δk is so small

$$e^{-k_{a_1} t} \approx e^{-k_{b_1} t}(1 - \Delta k t)$$

$$B = \frac{k_a[A]_0}{k_b - (k_{b_1} + \Delta k)}(e^{-k_{b_1} t}(1 - \Delta k t) - e^{-k_b t})$$

$$B = \frac{k_a[A]_0}{-\Delta k}(-\Delta k t e^{-k_b t})$$

$$B = k_a[A]_0(t e^{-k_b t})$$

$$\frac{dB}{dt} = k_a[A]_0(e^{-k_b t} - k_b e^{-k_b t} t) = 0$$

$$k_a[A]_0 e^{-k_b t}(1 - k_b t) = 0$$

$$(1 - k_b t) = 0$$

$$t_{max} = \frac{1}{k_b}$$

$$k$$

's are equal as the limit is approached. So for k = 0.1:

$$t_{max} = 10\ s$$

$$B_{max} = 0.3679\ \text{mol/L}$$

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